Organization Maturity Model

A Unified Framework for B2B SaaS Health Metrics

Security | Operational Excellence | Quality | Product | AI

Executive Overview

The Challenge

Fragmented Metrics Across the Organization

- **Security** tracks vulnerabilities, compliance, threat detection
- **Engineering** tracks DORA metrics, SLOs, incidents
- **Quality** tracks coverage, defects, test results
- **Product** tracks adoption, activation, churn
- **AI** tracks adoption, productivity, governance

Result: No unified view of organizational health

The Impact

What Fragmentation Costs Us

Problem	Impact
Siloed dashboards	Leaders see different pictures
Inconsistent maturity	Some areas advanced, others neglected
Unclear accountability	Who owns cross-cutting metrics?
Competing priorities	No framework for trade-offs
Reactive posture	Firefighting instead of improving

The Solution

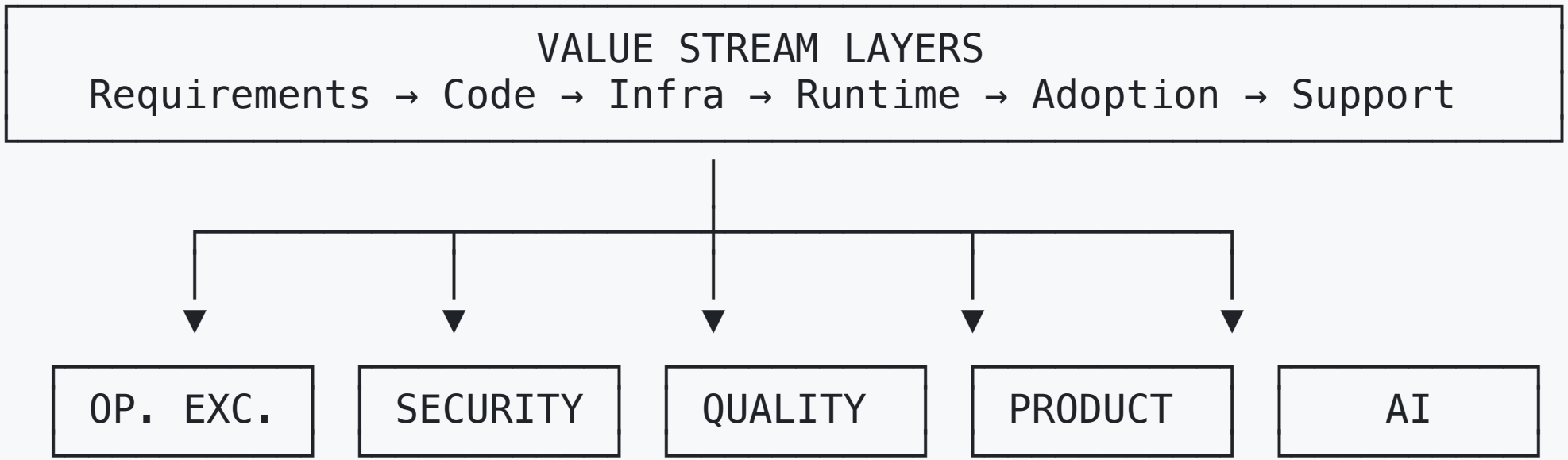
One Framework, Unified View

This maturity model provides:

- **Unified metrics** across all domains
- **Maturity levels** to measure capability
- **Clear accountability** through team ownership
- **Progression strategies** to guide improvement
- **Roadmap** to track advancement

Three Dimensions

Value Stream, Domains, and Lifecycle



Value Stream Layers

From Ideation to Support

Layer	Description	Owner
Requirements	Product ideation, specs, design	Product
Code	Application code, libraries, dependencies	Engineering
Infra	Cloud resources, networking, platform	Platform
Runtime	Running services, production workloads	SRE
Adoption	Product analytics, user engagement	Product/Growth
Support	Customer support, incident management	Support

Five Domains

Functional Areas with Standards

Domain	Focus	Key Metrics
Operational Excellence	Reliability, efficiency	DORA, SLOs, golden signals
Security	Protection, compliance	Vulnerabilities, MTTR, compliance
Quality	Testing, defects	Coverage, defect density, escape rate
Product	Adoption, retention	Activation, churn, NPS
AI	AI integration	Adoption, productivity, governance

Maturity Model

5 Levels of Capability

Level	Name	Description
M1	Reactive	Ad-hoc, firefighting, heroics
M2	Basic	Documented, some repeatability
M3	Defined	Standardized, consistent execution
M4	Managed	Data-driven, measured, controlled
M5	Optimizing	Continuous improvement, automated

Maturity Examples

What Each Level Looks Like

Level	Operational Excellence	Security	Quality	AI
M1	No SLOs, manual deploys	Ad-hoc scanning	No automation	Ad-hoc usage
M2	Basic monitoring	Scheduled scans	Some unit tests	Approved tools
M3	SLOs defined, CI/CD	Integrated scanning	Test automation	Prompt libraries
M4	Error budgets, DORA	Real-time detection	Quality gates	ROI tracked

Current State Summary

Where We Are Today

Domain	Current	Target	Gap
Security	M2	M4	CI integration, SIEM
Operational Excellence	M3	M4	Error budgets, DORA
Quality	M2	M4	Coverage, E2E tests
Product	M2	M4	Analytics, A/B testing
AI	M2	M4	Prompt library, ROI

Note: Replace with actual assessment data

Progression Strategies

How Should We Advance?

Strategy	Approach	Best For
Spike	One area to M5, others M2	Competitive differentiation
Wave	All together (M2→M3→M4→M5)	Balanced, low risk
Targeted	Priority to M4, others M3	Resource-constrained
Foundation	All to M3, then selective	Standardization first

Recommended Approach

Foundation + Targeted

Phase 1 (6 months): Establish foundation

- All areas to **M3** (Defined)
- Standardize processes, metrics, accountability

Phase 2 (6 months): Advance priorities

- Operational Excellence/Runtime to **M4**
- Security/Code to **M4**

Phase 3 (12 months): Optimize

- Selected areas to **M5**
- Based on business value

Roadmap Overview

Quarterly Progression

Quarter	Focus	Exit Criteria
Q1	Foundation	All domains M2→M3 in Code, Infra
Q2	Runtime	Operational Excellence/Runtime to M4
Q3	Security + AI	Security to M4, AI to M3
Q4	Optimization	Top 2 areas to M5

Security Domain

Prevention, Detection, and Response

Security Overview

What We Measure

The Security domain focuses on **protecting systems and data**:

- **Prevention** - Secure coding, vulnerability management
- **Detection** - Threat detection, anomaly identification
- **Response** - Incident response, remediation time

Goal: Shift security left while maintaining runtime protection

Security Across the Value Stream

Layer Responsibilities

Layer	Security Focus	Key Metrics
Requirements	Threat modeling, security requirements	Coverage %
Code	SAST, SCA, secrets scanning	Findings count, MTTR
Infra	Cloud security, IaC scanning	Misconfigurations
Runtime	DAST, WAF, runtime protection	Threats blocked

Security KPIs

Core Metrics by Maturity Level

Metric	M2	M3	M4	M5
SAST Coverage	≥50%	100%	100%	100%
Critical Vulns	Tracked	0 in prod	0 in prod	Auto-fixed
Secrets in Code	Scanned	0	0	0
MTTR (Critical)	Tracked	<14 days	<7 days	<1 day
Cloud Posture	-	≥80%	≥90%	≥95%
Detection Coverage	-	-	≥70%	≥90%

Security Maturity Levels

What Each Level Means

Level	Description	Indicators
M1	Reactive	Ad-hoc scanning, no process
M2	Basic	Scheduled scans, manual review
M3	Defined	Integrated scanning, policies enforced
M4	Managed	Real-time detection, automated response
M5	Optimizing	Predictive security, auto-remediation

Security Level Requirements

M2 → M3 → M4

M2 (Basic):

- SAST/SCA tools deployed
- Vulnerabilities tracked
- Incident process documented

M3 (Defined):

- 100% repos scanned in CI
- Security gates block critical findings
- 0 secrets in code

M4 (Managed):

Security Current State

Assessment Summary

KPI	Current	Level	M3 Target	M4 Target
SAST Coverage	75%	M2	100%	100%
Critical Vulns	5	M2	0 in prod	0 in prod
Secrets in Code	3	M2	0	0
MTTR (Critical)	14 days	M2	<14 days	<7 days
Cloud Posture	72%	M2	≥80%	≥90%

Security Roadmap

Prioritized Initiatives

Project	Impact	Timeline
Pipeline security gates	M2→M3 Code	Q1
Cloud posture management	M2→M4 Infra	Q1
SIEM/SOAR integration	M3→M4 Runtime	Q2
Secrets scanning	M3 Code	Q1
Auto-remediation	M4→M5 Code	Q3

Operational Excellence Domain

Reliability, Efficiency, and Performance

Operational Excellence Overview

What We Measure

The Operational Excellence domain focuses on **system reliability and delivery efficiency**:

- **Reliability** - Availability, latency, error rates
- **Efficiency** - Deployment frequency, lead time, resource utilization
- **Response** - Incident detection, recovery, post-mortems

Goal: Deliver reliable services efficiently with rapid recovery

Operational Excellence Across Value Stream

Layer Responsibilities

Layer	Focus	Key Metrics
Code	Build quality, CI/CD	Build success rate, lead time
Infra	Platform reliability	Resource utilization, drift
Runtime	Service health	Availability, latency, errors
Support	Incident management	MTTR, escalation rate

DORA Metrics

Mapped to Maturity Levels

Metric	M2	M3	M4	M5 (Elite)
Deploy Frequency	Monthly	Weekly	Daily	On-demand
Lead Time	1 month	1 week	<1 week	<1 hour
MTTR	<1 week	<1 day	<4 hours	<1 hour
Change Failure Rate	<45%	<30%	<15%	<5%

SLOs and Error Budgets

Target-Based Reliability

Service	SLO	Current	Error Budget
API Gateway	99.95% availability	99.92%	-3 hours
Payments	99.99% availability	99.995%	+4 hours
User Service	P99 < 200ms	185ms	Within budget

Error Budget = Allowed failures before SLO breach

Operational Excellence Maturity

What Each Level Means

Level	Description	Indicators
M1	Reactive	No SLOs, manual deploys, firefighting
M2	Basic	Basic monitoring, some CI/CD, incident response
M3	Defined	SLOs defined, CI/CD standard, runbooks
M4	Managed	Error budgets, DORA tracked, auto-scaling
M5	Optimizing	Self-healing, <1h lead time, proactive

Operational Excellence Current State

Assessment Summary

KPI	Current	Level	M3	M4	M5
Deploy Frequency	2x/week	M3	Weekly	Daily	On-demand
Lead Time	5 days	M3	1 week	1 day	1 hour
MTTR	4 hours	M4	1 day	4 hours	1 hour
Change Failure Rate	18%	M4	≤30%	≤15%	≤5%
Availability	99.92%	M4	99.5%	99.9%	99.99%
SLO Coverage	100%	M3+	100%	100%	100%
Runbook Coverage	85%	M3	≥80%	≥95%	100%

Operational Excellence Roadmap

Prioritized Initiatives

Project	Impact	Timeline
SLO-based alerting	M3→M4 Runtime	Q1
Automated rollback	M3→M4 Runtime	Q1
DORA dashboard	M3→M4 Code	Q1
Runbook standardization	M2→M3 Support	Q2
Chaos engineering	M4→M5 Runtime	Q3

Quality Domain

Testing, Defects, and Quality Assurance

Quality Overview

What We Measure

The Quality domain focuses on **software quality and testing effectiveness**:

- **Coverage** - Test coverage across code and features
- **Defects** - Finding and fixing bugs early
- **Prevention** - Quality gates that prevent escapes

Goal: High-quality releases with minimal production defects

Quality Across the Value Stream

Layer Responsibilities

Layer	Quality Focus	Key Metrics
Requirements	Acceptance criteria, testability	Spec coverage
Code	Unit tests, static analysis	Code coverage
Infra	Infrastructure testing	IaC test coverage
Runtime	Production quality monitoring	Defect escape rate

Quality KPIs

Core Metrics

Metric	Description	Target
Code Coverage	% of code tested	>80%
Test Pass Rate	% of tests passing	>99%
Defect Density	Defects per KLOC	<0.5
Escape Rate	Defects found post-release	<5%
MTTR (Defects)	Time to fix defects	<3 days
Flaky Test Rate	Unreliable tests	<1%

Quality Maturity Levels

What Each Level Means

Level	Description	Indicators
M1	Reactive	Manual testing, no automation
M2	Basic	Some unit tests, basic CI
M3	Defined	Test automation, quality gates
M4	Managed	Coverage targets, defect tracking
M5	Optimizing	Continuous testing, predictive quality

Quality Current State

Assessment Summary

KPI	Current	Level	M3	M4	M5
Unit Test Coverage	68%	M2	≥70%	≥80%	≥90%
E2E Test Coverage	65%	M2	≥80%	≥90%	≥95%
Test Pass Rate	96%	M2	≥98%	≥99%	≥99.9%
Defect Density	0.8/KLOC	M3	≤1.0	≤0.5	≤0.2
Escape Rate	12%	M3	≤15%	≤10%	≤2%
Flaky Test Rate	3%	M3	≤5%	≤2%	≤1%
Quality Gates	80%	M2	100%	100%	100%

Quality Roadmap

Prioritized Initiatives

Project	Impact	Timeline
Coverage enforcement	M3→M4 Code	Q1
E2E test expansion	M2→M3 Runtime	Q1
Flaky test elimination	Quality reliability	Q1
Performance test automation	M3→M4 Test	Q2
Production quality monitoring	M2→M4 Runtime	Q2

Product Domain

Requirements, Adoption, and Customer Success

Product Overview

What We Measure

The Product domain spans **two value stream layers**:

- **Requirements** - Product ideation, specs, design clarity
- **Adoption** - User engagement, activation, retention

Goal: Ship the right features that customers adopt and love

Product Across the Value Stream

Layer Responsibilities

Layer	Product Focus	Key Metrics
Requirements	Spec clarity, design quality	Completion rate, clarity score
Adoption	User engagement, feature usage	Activation, retention, adoption
Support	Customer feedback loop	NPS, feedback volume

Product KPIs

Core Metrics

Metric	Description	Target
Spec Clarity	% with acceptance criteria	>95%
User Activation	% completing onboarding	>80%
Feature Adoption	% using new features (30d)	>60%
Time to Value	Days to first value	<7 days
30d Retention	% still active at 30 days	>85%
Churn Rate	Monthly churn	<3%

Product Maturity Levels

What Each Level Means

Level	Description	Indicators
M1	Reactive	No specs, no analytics, guessing
M2	Basic	Basic specs, some usage tracking
M3	Defined	Spec templates, analytics dashboards
M4	Managed	Data-driven decisions, experimentation
M5	Optimizing	Predictive analytics, continuous discovery

Product Current State

Assessment Summary

KPI	Current	Level	M3	M4	M5
Spec Clarity	65%	M2	≥90%	≥95%	≥99%
Activation Rate	55%	M2	≥60%	≥75%	≥85%
Feature Adoption	50%	M2	≥80%	≥90%	100%
30-Day Retention	72%	M3	tracked	≥80%	≥90%
Churn Rate	4.5%	M3	tracked	≤3%	≤2%
NPS	42	M3	tracked	≥50	≥70
Time to Value	14 days	M3	≤14 days	≤7 days	≤3 days

Product Roadmap

Prioritized Initiatives

Project	Impact	Timeline
Analytics implementation	M2→M3 Adoption	Q1
Spec template rollout	M2→M3 Requirements	Q1
Activation flow optimization	Activation +15%	Q1
A/B testing platform	M3→M4 Adoption	Q2
Customer health scoring	M4→M5 Adoption	Q3

AI Domain

AI Adoption and Integration

AI Overview

What We Measure

The AI domain focuses on **AI adoption across all value streams**:

- **Adoption** - Tool usage, training coverage
- **Productivity** - Efficiency gains, ROI
- **Governance** - Policy, quality, security

Goal: Accelerate productivity through responsible AI integration

AI Across the Value Stream

Layer Responsibilities

Layer	AI Focus	Key Metrics
Engineering	Code assistants, test generation	Developer adoption, productivity
Operations	AIOps, incident response	Auto-diagnosis rate
Security	AI-powered scanning	Detection coverage
Support	Ticket automation	Auto-resolution rate
Governance	Policy, ROI tracking	Compliance, spend

AI KPIs

Core Metrics

Metric	Description	Target
Tool Adoption	Teams using AI tools	80%
Training Coverage	Staff with AI training	80%
Coding Assistant Usage	Developers with AI assistants	80%
Productivity Gain	Measured improvement	>20%
AI Output Accuracy	Acceptance rate	>80%
ROI Tracked	Investment return measured	Yes

AI Maturity Levels

What Each Level Means

Level	Description	Indicators
M1	Reactive	Ad-hoc AI usage, no governance
M2	Basic	Approved tools, policy, basic training
M3	Defined	Prompt libraries, AI in CI/CD, quality guidelines
M4	Managed	ROI tracked, AIOps, AI security scanning
M5	Optimizing	Autonomous agents, custom models, self-healing

AI Level Requirements

M2 → M3 → M4

M2 (Basic):

- AI tools approved and deployed
- Usage policy documented
- 50% staff trained
- Spend tracked

M3 (Defined):

- 80% developers with AI assistants
- Prompt/template library established
- AI code review in 50% PRs

AI Current State

Assessment Summary

KPI	Current	Level	M3	M4	M5
Tool Adoption	60%	M2	≥80%	≥90%	≥95%
Training Coverage	45%	M1	≥80%	≥90%	≥95%
Coding Assistant	55%	M2	≥80%	≥90%	≥95%
Prompt Library	No	M2	Yes	Yes	Yes
AI Code Review	20%	M2	≥50%	≥80%	≥95%
Productivity Gain	15%	M3	-	≥20%	≥40%
ROI Tracked	No	M3	-	Yes	Yes

AI Roadmap

Prioritized Initiatives

Project	Impact	Timeline
Roll out AI coding assistants	M2→M3 Engineering	Q1
Create prompt library	M3 Enablement	Q1
AI training program	M2 Foundation	Q1
Integrate AI in CI/CD	M3 Engineering	Q2
Deploy AI metrics dashboard	M3→M4 Governance	Q2
AIOps implementation	M4 Operations	Q3

Summary and Next Steps

Cross-Domain Summary

Maturity Progression Plan

Domain	Current	Q2 Target	Q4 Target
Security	M2	M3	M4
Operational Excellence	M3	M4	M4
Quality	M2	M3	M4
Product	M2	M3	M4
AI	M2	M3	M4

Investment Required

Resources and Commitment

Area	Investment	Purpose
Tooling	Platform improvements	Automation, observability, AI
Training	Team enablement	SRE, security, quality, AI practices
Headcount	Enabling team	Cross-functional improvement
Time	Team allocation	20% on maturity initiatives

Success Metrics

How We'll Measure Progress

Metric	Current	6 Month	12 Month
Average Maturity	M2.4	M3.0	M3.5
SLO Compliance	85%	95%	99%
MTTR	4 hours	2 hours	1 hour
Deployment Frequency	Weekly	Daily	On-demand
AI Productivity Gain	15%	25%	40%

Next Steps

Immediate Actions

1. **Approve** maturity model as the unified metrics framework
2. **Conduct** baseline maturity assessment per domain
3. **Select** progression strategy (recommend: Foundation + Targeted)
4. **Assign** domain overlay owners
5. **Kickoff** Q1 initiatives

Discussion

Questions?

Key Decisions Needed:

- [ ] Approve maturity framework adoption
- [ ] Select progression strategy
- [ ] Allocate resources for Q1 initiatives
- [ ] Assign executive sponsors per domain

Appendix

Team Accountability

Team Topologies Integration

Team Type	Role	Accountability
Stream-Aligned	Build and run services	Layer metrics (code, runtime)
Platform	Provide infrastructure	Infra layer metrics
Enabling	Help teams adopt practices	Cross-team improvement
Overlay	Define domain standards	Domain-wide metrics

Every metric has an owner. Every team knows their KPIs.

Framework Mappings

Industry Alignment

Framework	Mapping
DORA	Operational Excellence metrics
SRE	Golden signals (latency, traffic, errors, saturation)
NIST CSF	Security domain stages
MITRE ATT&CK	Security detection metrics
ISO 25010	Quality characteristics

Project Prioritization

Scoring Framework

Factor	Weight	Description
KPI Impact	30%	Expected metric improvement
Maturity	25%	Enables level progression
Business Value	20%	Revenue, risk, compliance
Effort	15%	Time and resources (inverse)
Dependencies	10%	Blockers (inverse)

Projects scored 1-5 on each factor, weighted total determines priority.

Glossary

Term	Definition
SLO	Service Level Objective - target for a metric
SLI	Service Level Indicator - the measurement
DORA	DevOps Research and Assessment metrics
MTTR	Mean Time to Recovery
CFR	Change Failure Rate
AIOps	AI for IT Operations
CSPM	Cloud Security Posture Management